



SOFIENE HAMZAOU

AI/LLM Engineer | Full Stack Developer | Security Engineer | Professional Truck Driver

admin@vaultguard.eu.org +216 22 090 800
sniper4u2.github.io vaultguard.eu.org

PROFESSIONAL SUMMARY

AI/LLM Engineer, Full Stack Developer, and Security Engineer with 6 years of specialized experience in machine learning, cybersecurity, and systems architecture. 20 years of technical experience with computers and software. Since 2018, focused on ethical hacking, security tools, shell scripting, and full-stack development (8 years cybersecurity). Since 2020, specialized in AI/LLM engineering, fine-tuning, and SFT dataset curation (6 years AI/LLM). Also experienced as a professional heavy vehicle driver (C+E license) with 3 years of commercial transport operations.

TECHNICAL SKILLS - AI & MACHINE LEARNING

- **Programming Languages:** Python, TypeScript, JavaScript, Objective-C, SQL
- **AI/ML:** LLM Fine-Tuning, SFT Datasets, Neuro-Symbolic AI, PyTorch, Hugging Face Transformers
- **RAG & Vector Databases:** ChromaDB, Embeddings, Semantic Search
- **Model Quantization:** GGUF, llama.cpp, CUDA GPU offloading
- **Agent Frameworks:** smolagents, CodeAgent, Gradio UI
- **Backend:** FastAPI, Node.js, Cloudflare Workers, Docker, Redis, PostgreSQL

- **Frontend:** React, Vite, HTML/CSS

TECHNICAL SKILLS - CYBERSECURITY

- **Penetration Testing:** Red Team Operations, Vulnerability Assessment
- **Systems Forensics:** Incident Response, Malware Analysis
- **CVE Analysis:** Security Research, Threat Intelligence
- **Authentication:** WebAuthn/FIDO2, JWT, AES-256-GCM encryption
- **SS7/GSM:** Telecom Protocol Security (for authorized security research)
- **Infrastructure:** Cloudflare, Supabase, Linux, Git/GitHub

TECHNICAL SKILLS - HEAVY VEHICLE OPERATIONS

- **License Class:** C+E (Heavy Vehicles with Trailers)
- **Vehicle Types:** Semi-remorque trucks, articulated trucks, commercial transport
- **Vocational Training:** 1.5 years certified professional training
- **Operations:** Logistics, long-haul transport, route planning, load management

KEY PROJECTS - AI & MACHINE LEARNING

VaultGuard Institutional P2P Platform

- Institutional-grade crypto security and OTC settlement engine
- 137 API endpoints on Cloudflare Edge infrastructure
- Zero-Knowledge Proof identity disclosure system
- WebAuthn/FIDO2 passwordless biometric authentication
- Full-stack: React frontend, TypeScript backend, Supabase database

- Live URL: vaultguard.eu.org

B13 Cybersecurity SFT Dataset Collection

- 16,874 unified ChatML SFT training samples
- 10,000 technical CVEs with Chain-of-Thought reasoning
- 130,173 security evaluation records
- Published on Hugging Face ([sofienehmz/b13-cybersecurity-sft-datasets](https://huggingface.co/sofienehmz/b13-cybersecurity-sft-datasets))
- Gated access for security reasons (contains 100k+ working CVEs)

HSPN-MAI ARC-AGI Neuro-Symbolic Synthesis

- Kaggle ARC Prize 2026 competition entry
- Ranked 176 of 3,885 teams (score 30.45)
- Neuro-Symbolic AI combining neural networks with program synthesis
- Authored 10-page formal research treatise on out-of-distribution generalization
- Code private for ongoing competition

WebDriverAgent iOS Automation Framework

- Complete WebDriverAgent v11.1.2 implementation for iOS 26.1
- Auto-built via GitHub Actions CI/CD (1,575 commits)
- Comprehensive test suite with mocha testing framework
- Custom build scripts and Fastlane integration
- GitHub: github.com/spaypeur/WebDriverAgent

Sports Betting Platform (Full-Stack)

- Full-stack application with React/Vite frontend
- Supabase PostgreSQL database with edge functions
- Cloudflare Workers for serverless API architecture
- Real-time odds integration with multiple sports APIs
- Development paused due to client payment dispute

C2 Server Framework

- Advanced Command & Control framework with FastAPI and WebSockets
- Autonomous AI pentesting agent orchestration
- Dynamic CVE threat intelligence mapping
- SS7/GSM exploitation capabilities (for authorized security research)
- Comprehensive CVE exploitation database
- Private repository - access restricted due to sophisticated capabilities

Zindi Competition Projects

- **Barbados OCR:** Multi-architecture HTR system with PyTorch and Kraken
 - Competition: R.O.A.D. Barbados Historic Handwriting Challenge
 - Team: Tunisian(x B13) - Rank 292 (Public Score: 0.877, Private Score: 0.879)
 - URL: zindi.world/competitions/road-barbados-historic-handwriting-challenge
- **Bias Bounty Equity:** Algorithmic fairness pipeline with geospatial features
 - Competition: Bias Bounty Mapping Equity
 - URL: zindi.world/competitions/bias-bounty-mapping-equity

- Both projects: Local development for competition entries

Tutti Phone Extractor

- Web scraping tool for automotive platform listings
- Developed for client project
- Discontinued after discovering client intended to intercept listings before publication
- GitHub: github.com/spaypeur/tutti-phone-extractor

PROFESSIONAL EXPERIENCE

Independent Technical Consultant

2018 - Present

- AI/LLM Engineering: Fine-tuning models, building SFT datasets, creating RAG systems
- Cybersecurity: Penetration testing, security auditing, vulnerability assessment
- Full-Stack Development: Building web applications, APIs, and backend systems
- Systems Architecture: Designing secure, scalable infrastructure
- Client work across Europe and USA

Professional Heavy Vehicle Driver

2012 - 2015

- 3 years commercial heavy vehicle driving experience
- License Class C+E (vehicles with trailers)
- Semi-remorque and articulated truck operations
- Long-haul transport, logistics coordination, route planning
- Vocational Training Certificate (1.5 years certified training)
- Available for truck driving positions in EU and USA

Client Services Coordinator

2008 - 2011

- International hotels in Tunisia and Turkey
- Caribbean World Monastir, Thalasso Monastir, Ksar Kervansaray (Tunisia)
- 5-star hotel in Bodrum, Turkey
- Client relations, event coordination, multilingual daily operations
- Languages used: Arabic, French, English, Russian, German

EDUCATION

Pharmaceutical Studies in Russia

2013 - 2016

- 1 year Russian language learning (B2 level)
- 3 years pharmacy studies at Perm State Pharmacy Academy, Russia
- Ryazan State Medical University: Pharmacy program (2nd year)

Civil Aviation Training

2008

- Flight Dispatcher Diploma N°228/2008 (Ministry of Transport, Tunisia)
- CMFC Personnel Navigant Commercial (Cabin Crew)

Vocational Training

2010 - 2011

- Heavy Vehicle C+E License (1.5 years certified training)
- Professional heavy vehicle driving certification

Baccalaureate Diploma

2007

- Arts Baccalaureate (Tunisia)

CERTIFICATIONS

AI & Machine Learning

- HuggingFace AI Agents Course
- HuggingFace / DeepLearning.AI Fundamentals of LLMs

Cybersecurity (IBM SkillsBuild)

- IBM Incident Response & Systems Forensics
- IBM Security Operations & Management
- IBM Vulnerability Management
- IBM System & Network Security
- IBM Cloud Security
- IBM GRC & Data Privacy
- IBM Cybersecurity Fluency Pathway
- IBM Getting Started with Cybersecurity

Software Engineering

- Microsoft / freeCodeCamp Foundational C#
- freeCodeCamp Scientific Computing with Python
- freeCodeCamp Frontend Dev Libraries V8
- freeCodeCamp Web Design Certification V8
- freeCodeCamp Legacy Frontend Development

Language & Aviation

- IELTS General Training Band 5.5 / B2 (British Council)
- Civil Aviation Flight Dispatcher Diploma N°228/2008
- CMFC Personnel Navigant Commercial (Cabin Crew)

Vocational

- Heavy Vehicle C+E License

- Driving Vehicles with Trailers License

LANGUAGES

- Arabic: Native
- French: Fluent
- English: Professional (IELTS B2)
- Russian: B2 (1 year language study + 3 years university in Russia)
- German: Good
- Dutch: Basic
- Italian: Basic

INTERESTS & HOBBIES

- Archaeological field research and historical site exploration
- Roman, Punic, and Islamic-era archaeology
- Ancient Mediterranean civilizations
- Cultural heritage documentation

CONTACT

- Email: admin@vaultguard.eu.org
- Phone: +216 22 090 800
- GitHub: github.com/sniper4u2 | github.com/spaypeur
- Hugging Face: huggingface.co/sofienehmz
- VaultGuard: vaultguard.eu.org
- Portfolio: sniper4u2.github.io